

AI POWERED INTERVIEW PREPARATION AND RESUME COACH

PHASE I REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled “**AI POWERED INTERVIEW PREPARATION AND RESUME COACH**” is the bonafide work of **KAVEESHAA S (230701146), KAVISH RAGAV S (230701147), KEERTHIVARMAN AB (230701151), KRITHIKA PV (230701157)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

“This project addresses the following Sustainable Development Goal(s):
SDG 4 – Quality Education, SDG 8 – Decent Work and Economic Growth”

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ABSTRACT

The AI-Powered Interview Preparation & Resume Coach is a web-based platform that helps students prepare for placements through a single integrated system. It combines resume analysis, ATS optimization, AI-powered mock interviews, coding practice, and performance tracking.

The system analyzes resumes and compares them with job descriptions to identify matching and missing skills. It also conducts role-specific mock interviews and evaluates responses based on accuracy, clarity, grammar, and confidence. Coding practice and real-time coding battles help students improve their programming and problem-solving skills.

The platform tracks performance, identifies skill gaps, and provides personalized feedback to improve overall preparation. The main objective is to reduce the need for multiple disconnected platforms and help students become more confident, skilled, and job-ready.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ATS	Applicant Tracking System
API	Application Programming Interface
DB	Database
DOCX	Microsoft Word Document Format
HTTP	HyperText Transfer Protocol
JWT	JSON Web Token
LLM	Large Language Model
NLP	Natural Language Processing
PDF	Portable Document Format
REST	Representational State Transfer
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator
JD	Job Description
MERN	MongoDB, Express, React, Node.js

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Getting ready for placements is an important part of a student's academic journey. Students are expected to prepare for several stages of recruitment, including resume screening, aptitude tests, coding assessments, technical interviews, HR interviews, and communication rounds. Preparing for all these stages at the same time can be difficult, especially when different platforms are required for each activity.

The AI-Powered Interview Preparation & Resume Coach is a unified web-based platform designed to make placement preparation easier and more organized. The system brings resume analysis, ATS evaluation, job matching, AI-powered interviews, coding practice, aptitude assessment, communication evaluation, skill recommendations, and performance tracking into one platform.

The main idea of the project is to provide students with a single environment where they can understand their current level of preparation, identify their weak areas, and work on improving them. Instead of providing only a score, the system attempts to give useful feedback and recommendations based on the student's performance.

The project focuses on making placement preparation more personalized and less fragmented. It allows students to repeatedly practice different recruitment activities and track their progress over time.

The project objectives and key capabilities are based on the First Review presentation.

1.2 OBJECTIVE

The primary goal of the project is to create a centralized, intelligent platform that can help students enhance their readiness for jobs. The system will aim at analyzing the resume of the candidate based on the content, structure, skills, experience, projects, and ATS readability, along with the ATS score and areas to improve on. Moreover, it will compare the resume of the candidate against certain jobs to determine the skills required for that particular job description. This will include role-specific mock interviews using artificial intelligence (AI). The interviews will be evaluated on the basis of relevance, correctness, clarity, grammar, and confidence. Furthermore, there will be practice coding to enhance programming skills and solve problems, aptitude test for the purpose of placement preparation, and communication assessment. The system will also identify skills gaps for the individual users.

1.3 EXISTING SYSTEM

A number of platforms already exist for individuals to perform preparation activities for placements. Platforms that deal specifically with resume creation include Canva and Zety. Interviewing platforms include Pramp and InterviewBit. Coding platforms like LeetCode and HackerRank emphasize programming tests and coding skills. Lastly, platforms such as IndiaBIX offer aptitude training. However, these platforms only help with one aspect of placement preparations. Thus, a

student will need to shift from one platform to another in order to prepare properly. The current process makes the preparation process fragmented because resume skills, coding skills, interviewing skills, aptitude skills, and communication skills are not related to each other.

1.4 PROPOSED SYSTEM

This solution consists of many activities for placement preparation under one intelligent web-based application. This application will allow students to submit and analyze their resume, check ATS compatibility, and compare their resume with certain job descriptions. Students also can practice with mock interviews, technical interviews, code challenges, aptitude tests, and communication tasks. Results of performance in these tasks are then analyzed to recognize skill gaps and weaknesses. Based on these weaknesses, appropriate recommendations and learning paths are recommended by this system. In general, workflow of the system consists of the following activities: Resume or Assessment → Analysis → Performance Tracking → Skill Gap Recognition → Recommendations → Learning Paths → Improvement. Thus, by incorporating all these activities into one web application, the limitations of fragmented placement preparation are overcome.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Artificial Intelligence and NLP have been gaining their value in recruiting and employability-related applications. The literature has covered such areas as automated communication evaluation, resume ranking, resume-job matching, AI interviews, programming feedback, job recommendation, and automated resume screening.

Pereira, Meng, and Hone (2021)

The research concerned prediction of communication effectiveness through automatic non-verbal recognition technologies. Machine learning classifiers were used for analysis of facial expressions, hand movements, vocal behavior, and other types of non-verbal communication.

It was shown that automated analysis may be helpful for communication skills evaluation. The Naive Bayes classifier demonstrated 78% of prediction accuracy. This article is related to the communication evaluation aspect of the proposed solution.

Tejaswini K., Umadevi V., Kadiwal and Revanna (2021)

The researchers proposed a machine-learning-based resume ranking system. TF-IDF was used to represent resumes and job descriptions, while cosine similarity and KNN were used for ranking.

The study reported 85% text-parsing accuracy and 92% ranking accuracy. This work supports the use of automated resume analysis and resume-job matching.

Rojas-Galeano, Posada and Ordoñez (2022)

This research studied AI-based job-resume matching and examined approaches involving NLP, information extraction, semantic representation, and matching.

The research shows the development of recruitment systems from traditional NLP methods toward embeddings, deep learning, and transformer-based approaches.

Enhanced DSSM Technique for Job Recommendation (2022)

This study used Deep Semantic Structure Modeling to represent job descriptions and skills using semantic embeddings.

The research demonstrated the usefulness of semantic representations for improving job recommendation and addressing scalability challenges.

Liu, Wei, Wu and Luo (2023)

This study investigated how job applicants communicate with AI interviewers. It compared different interview conditions involving human and AI evaluation.

The research highlighted the importance of user experience, perceived uncertainty, social presence, and speech fluency when designing AI interview systems.

Kiesler, Lohr and Keuning (2023)

The study explored the use of Large Language Models for generating formative programming feedback.

The research showed that LLMs can provide useful programming feedback, but misleading or incorrect suggestions can still occur.

Gan, Zhang and Mori (2024)

The researchers proposed an LLM-agent framework for automated resume screening. The framework included resume extraction, classification, summarization, grading, and recruitment decision support. The study reported an F1 score of 87.73% for sentence classification.

Nagasawa, Okada, Ishihara and Nitta (2024)

The study proposed an adaptive interview strategy based on interviewee speaking willingness. Multimodal features and Random Forest classification were used to select interview questions adaptively. The study demonstrated that adaptive questioning can improve interaction in AI-based interviews.

Rai, Abhiram, Adithya and Hrithik (2024)

The researchers proposed an AI-based interview evaluator using computer vision, speech recognition, and confidence classification. The research demonstrates the possibility of combining speech, emotion, and confidence analysis for automated interview evaluation.

Jacobs and Jaschke (2024)

The research evaluated the use of GPT-4 for programming feedback. The results showed that LLM-generated feedback can be useful for coding errors, although hallucinated or incorrect suggestions remain a limitation.

Wilson and Caliskan (2024)

The study investigated gender, race, and intersectional bias in AI-based resume screening. The research highlights the importance of fairness and bias awareness when designing automated recruitment systems.

Guan et al. (2024)

The JobFormer study proposed a skill-aware job recommendation system using semantic-enhanced transformers.

The study demonstrates the importance of skill-aware representations in personalized job recommendations.

Lund et al. (2025)

Resume2Vec explored transformer-based resume embeddings for candidate matching.

The study demonstrates the potential of embedding-based approaches to improve resume-job matching beyond traditional keyword matching.

2.2 COMPARISON AND DISCUSSION

There is plenty of evidence in the literature on how AI may be used for individual placement and recruitment.

The research on resumes is concentrated on information extraction, candidate ranking, and resume-job matching. The research on interviews centers on automated assessment of communication, confidence, speaking and answering skills. Research on programming considers the application of LLMs to provide feedback on coding activities.

Still, these techniques are mostly focused on a particular recruitment activity but not on creating an all-around preparation environment for students.

Our system integrates these aspects into one platform. Instead of considering resume writing, interviews, coding, aptitude and communication as independent tasks, we link them by performance

monitoring and skill recommendations.

2.3 CONCLUSION

Literature survey results indicate that AI and NLP can be used efficiently for analyzing resumes, matching jobs, interviews, giving feedback on programming skills, and recommendations. Some limitations identified in the literature include wrong feedback from AI systems, biases, and user experience. As a result, the proposed system is intended to provide an integrated placement preparation system while making evaluation and recommendations clear for users.

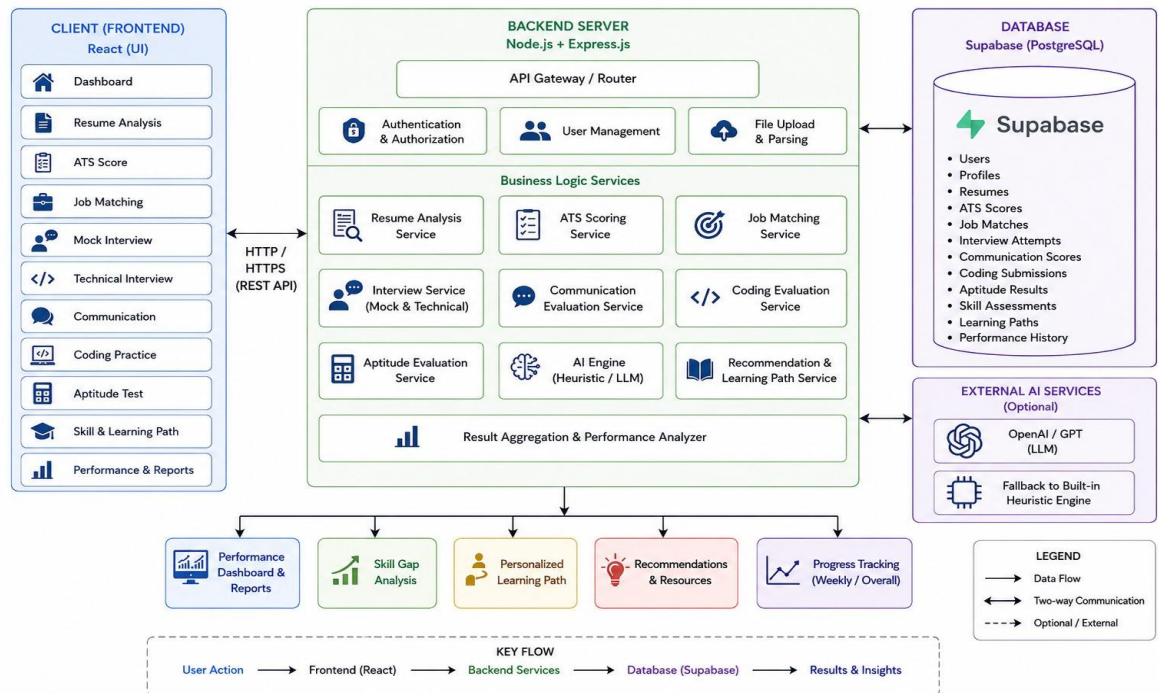
CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

This system is based on the client-server architecture over the web. The frontend serves the user interface through which the student can interact with the various modules of placement preparation. The communication between frontend and backend takes place through REST APIs. The backend controls authentication, resume processing, ATS evaluation, interview assessment, coding, aptitude testing, recommendation, and performance monitoring. The application uses Supabase PostgreSQL database to store user information, resumes, interview attempts, coding attempts, aptitude scores, communication scores, learning tracks, and skill tests.

Figure 3.1. Architecture Diagram of the System



3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

The software technologies used in the project are:

COMPONENT	TECHNOLOGY
Frontend	React
Build Tool	Vite
Styling	Tailwind CSS
Backend	Node.js
Server Framework	Express
Database	Supabase PostgreSQL
Authentication	JWT and bcryptjs
API Communication	Axios
File Upload	Multer
PDF Processing	pdf-parse
DOCX Processing	mammoth
Charts	Recharts
Security	Helmet and express-rate-limit

Project presentation sees the solution as a single web based application, while the implementation consists of frontend, backend, database and AI services.

3.2.2 HARDWARE REQUIREMENTS

The system can be developed and used on a standard modern computer.

COMPONENT	REQUIREMENT
Processor	Intel Core i5 or equivalent
RAM	8 GB or above
Storage	256 GB or above
Display	1366 × 768 or above
Network	Internet connection
Input	Keyboard and Mouse
Browser	Modern Web Browser

3.2.3 DATASET REQUIREMENTS

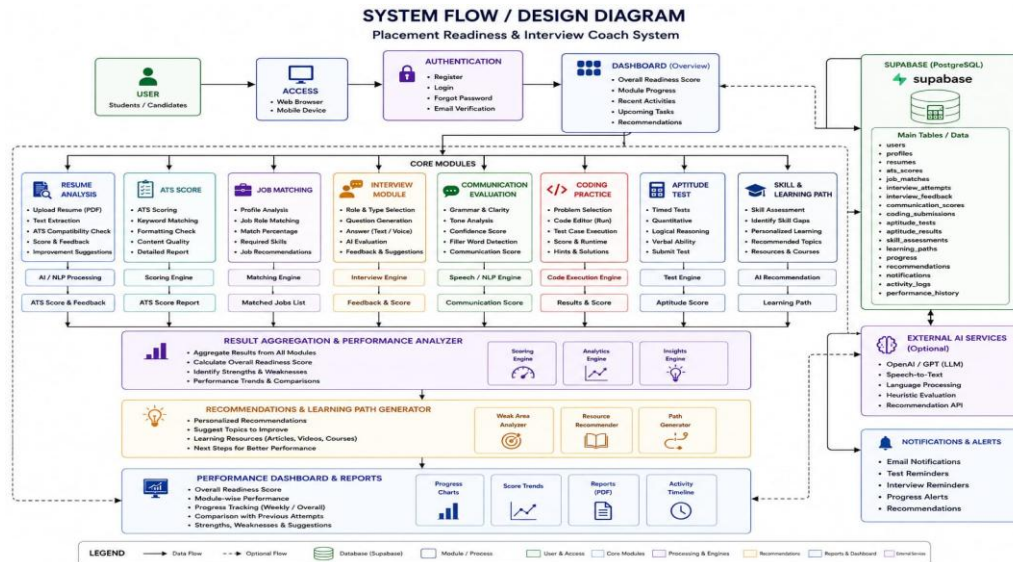
This project does not rely on the use of a big central machine learning dataset to carry out the process of the main application. Rather, the system relies heavily on the user-generated data which includes resumes, job description, interview questions and answers, coding questions and answers, aptitude questions and answers, communication responses and performance of the user. Literature survey acts as the external research basis for AI recruitment and assessment, whereas application uses user-generated data during actual usage.

3.2.4 DEPLOYMENT AND SCALING

The design of the system has been created using separate frontend, backend, service, and database modules. This helps in creating an easy to use system that is easy to scale and maintain. The frontend can be used as a web app, whereas the backend can be used as a Node.js service with integration of Supabase PostgreSQL as the persistent database. The modularity of the system has made it possible to create future services like LLN evaluation, enhanced semantic match, more programming languages, etc.

3.3 SYSTEM DESIGN

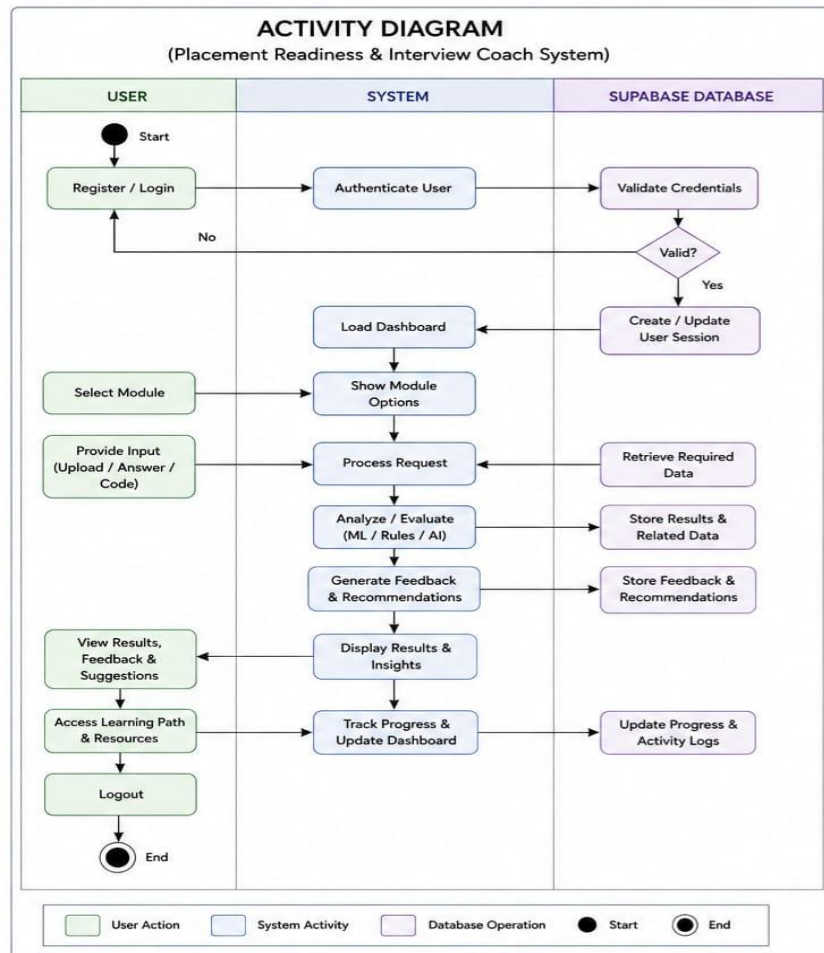
The system is designed to link all of the placement preparation actions through the user's performance record and profile.



3.3.1 ACTIVITY DIAGRAM

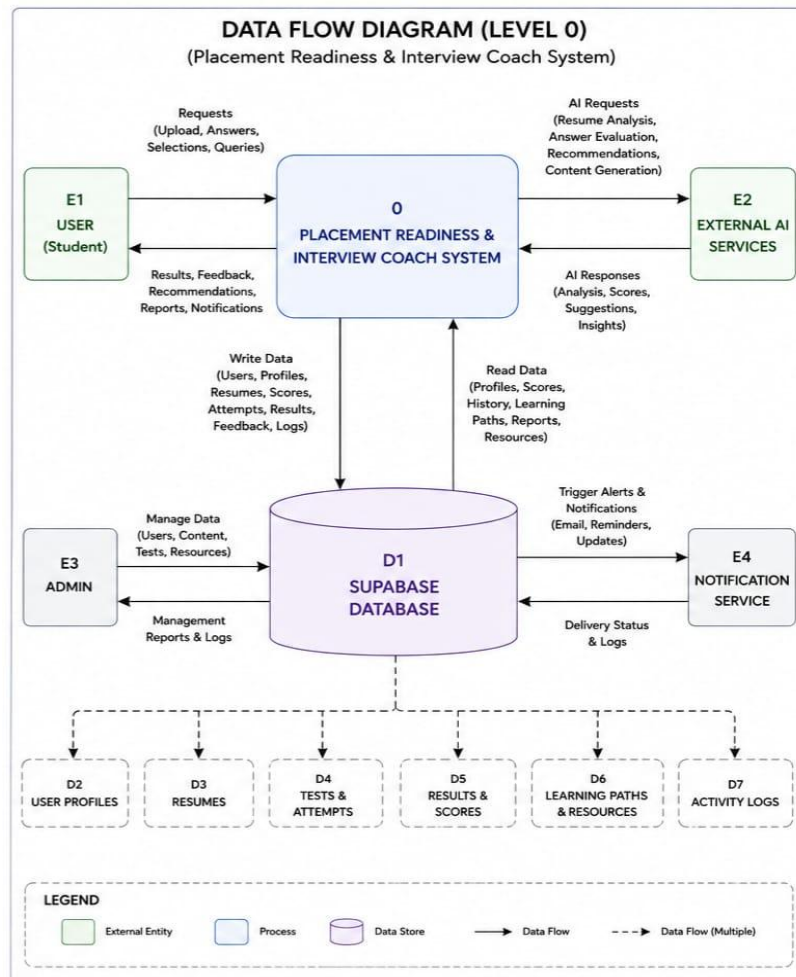
This activity diagram depicts the process of actions undertaken by the student when using the platform. First, the user registers

himself/herself or logs into the system. Next, after successful authentication, the user uploads his/her resume, conducts ATS analysis, matches his/her resume to the job description, participates in interviews, performs coding exercises, undertakes aptitude tests, and analyzes communication. These results are analyzed and stored. Finally, based on performance data collected throughout this process, skill gaps are identified and recommendations are made.



3.3.2 DATA FLOW DIAGRAM

Data Flow Diagram (DFD) shows how information flows from the user, application modules, artificial intelligence services, and to the database. User inputs include resumes, job descriptions, interview answers, source codes, and aptitude test answers. Application will process this data and produce scores, comments, recommendations, and learning paths. The output data will be saved in the database and then shown on the performance dashboard.



CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

This project uses a modular and service-oriented approach to combine different placement preparation activities into one platform. Within Resume Analysis Methodology, the resume uploaded by the user is initially processed to obtain text data. This data is then analyzed to determine the key parts of the resume, skills, experience, projects, contact details, and other related keywords. Within ATS Analysis Methodology, the resume is scored depending on various aspects such as keyword matching, experience and projects, section completion, structure, language, and contact information.

Within Job Matching Methodology, the resume of the user is matched to the requirements of a selected job description to find out the matching and lacking skills of the user. It will help the user see whether his or her resume matches the target job position or not. Within Interview Evaluation Methodology, role-related questions are generated and the answers of the user are rated taking into account such aspects as relevance, clarity, grammar, confidence level, and technicality. The results obtained from various modules are stored within Performance Analysis Methodology and used to define the weak points of the user. Recommendations for the user are provided by the Recommendation Methodology using the discovered skills gaps.

4.2 MODULES

The system is divided into eight major modules to keep the application organized and easy to maintain.

4.2.1 USER AUTHENTICATION MODULE

The User Authentication Module is responsible for the management of accounts and user access.

Users can sign up by providing their mandatory details. Existing users can log in through their credentials. Users are also provided with profile management and restricted access to application modules. This helps to ensure that the resume data, assessments, interview records, and progress of the user are linked to the user.

4.2.2 RESUME ANALYSIS AND ATS MODULE

The Resume Analysis & ATS Module is one of the main modules of the system. It gives the users an opportunity to upload their resumes and get their automated analysis done by the system. The system will take all the required information out of the uploaded resume and analyze the content and structure of the resume. All the details like contact information, education, experience, projects, skills, certification, languages, keywords, and structure of the resume will be analyzed in the process. The ATS component of the system will provide the overall compatibility score of the resume and the areas which need improvement.

4.2.3 JOB MATCHING MODULE

The Job Matching Module matches the resume of the user with a particular job description to see how well the candidate matches the requirements of the selected job description. Relevant requirements of the selected job description are matched with the information present on the resume and provide information related to matching skills, missing skills, matching keywords, job requirements, match score, and recommendations. This allows students to have an insight into the skills that they possess and the skills that need to be improved in order to qualify for a certain job role.

4.2.4 AI INTERVIEW AND COMMUNICATION MODULE

The AI Interview and Communication Module allows users to take part in various role-based interviews without the involvement of a human interviewer in each of the sessions. There are different types of interviews that one can take part in and these include behavioral, technical, HR and case study interviews. The system judges the performance of the user depending on criteria such as relevancy, accuracy, clarity, grammar, confidence and technicality of the answer given. The communication aspect evaluates how effectively the person communicates his/her points.

4.2.5 CODING AND APTITUDE ASSESSMENT MODULE

This module is concerned with two key aspects of the preparation for placement: programming and aptitude. Through the programming aspect, students are able to engage in programming exercises and evaluate their solutions, which in turn helps in improving logical and problem-solving skills, programming, and technical interviews preparation. This

project also provides coding exercises and coding battles in real-time to make the preparation process more interactive. The aptitude aspect enables students to solve aptitude questions and evaluate their performance. Through this system, various important information such as scores, accuracy, correct answers, wrong answers, questions not answered, time taken, and weak areas are recorded.

4.2.6 SKILL RECOMMENDATION AND LEARNING PATH MODULE

The Skill Recommendation and Learning Path Module takes the result of the assessment and turns it into action plans for improvement. It evaluates the user's performance and determines the weak areas that need further preparations. If the user shows a poor performance in terms of communication skills or coding, it will determine the weak area and suggest the appropriate learning activities.

4.2.7 PERFORMANCE TRACKING AND DASHBOARD MODULE

The Performance Tracking and Dashboard Module is a tool that presents an overview of the student's placement preparation in one place. The dashboard shows useful data such as the student's total performance, performance in modules, level of skills, latest activity, weekly progress, progress on the learning path, and those areas that should be improved. The performance of various activities in preparation can be shown by using the dashboard in order to know one's present status without visiting each module.

4.2.8 ADMIN/STAFF MANAGEMENT MODULE

In the Admin/Staff Management Module, there exists an exclusive portal through which authorized staff members can track and manage the activities done by the students in preparing for their placement. The system has two separate interfaces for the students and the staff; thus, the staff members do not have to log into the student portal in order to access the administrative functionalities. The Staff Portal enables authorized staff members to track the performance of the students and their interview activities, and access necessary analytics.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

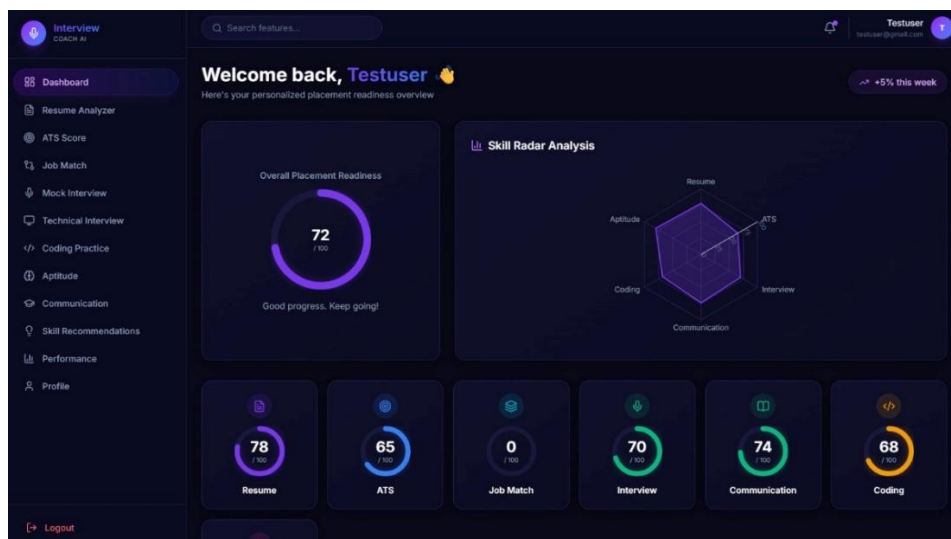
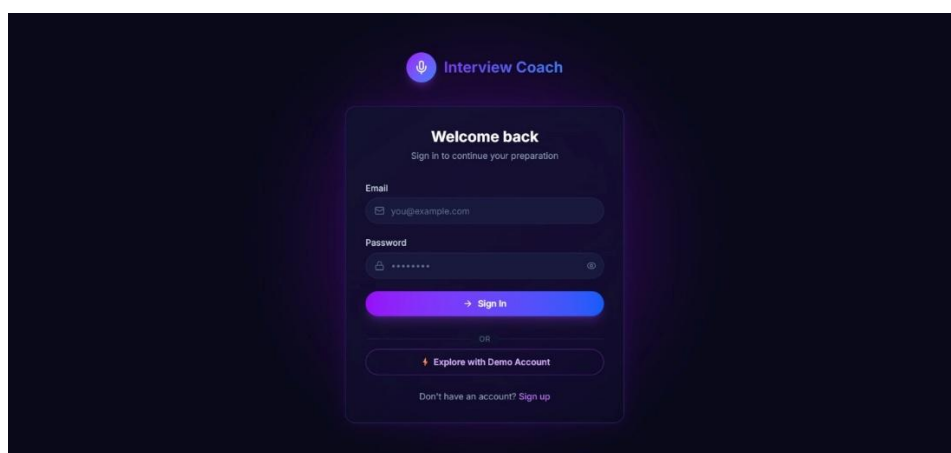
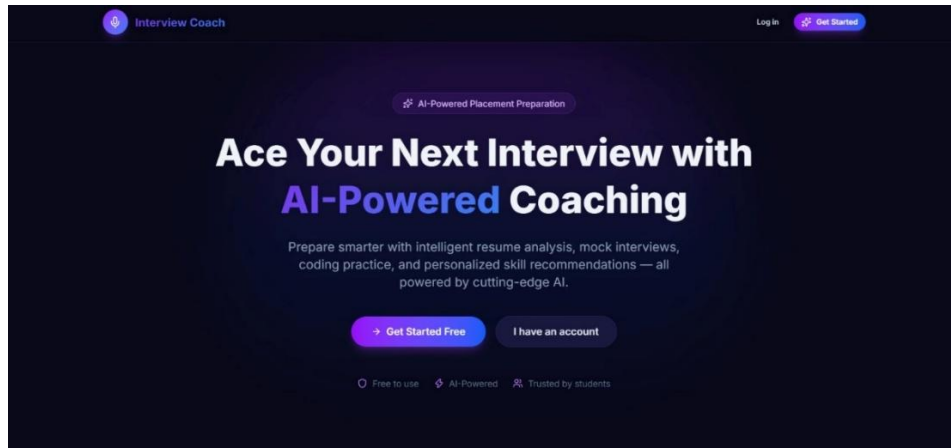
5.1 DEVELOPMENT ENVIRONMENT

The development project has been carried out as a web application having a frontend, a backend, a database, and an analysis part. The frontend is responsible for providing the user interface, and it uses REST APIs to communicate with the backend which, in turn, performs the task of application logic handling and database handling using Supabase PostgreSQL database. The main technologies that have been used for developing this application are React and Vite for frontend, Tailwind CSS for styling purposes, Node.js and Express for backend, Supabase PostgreSQL for the database, JWT and bcryptjs for authentication purposes, Axios for the communication between APIs, Multer for uploading files, pdf-parse and mammoth for parsing the resume documents, and Recharts for visualizing the performance data.

5.2 FRONTEND IMPLEMENTATION

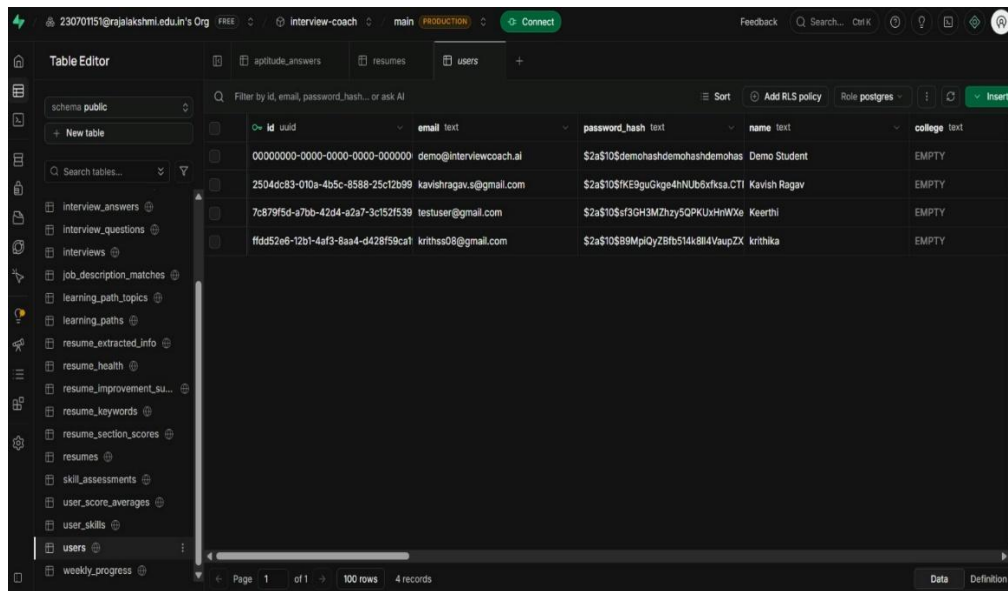
React is used to develop the frontend of the application based on the approach of components where commonly used UI elements can be shared across multiple pages. There are some of the important modules in the application which include: Login & Registration, Student Dashboard, Resume Analyzer, ATS Score, Job Matching, Mock Interview, Technical Interview, Communication Evaluation, Coding Practice, Aptitude Test, Skill Recommendations, Learning Path, Performance History, User Profile, and Admin/Staff Portal. Admin/Staff portal is the separate UI

section provided for the staff to keep track of the performance of the students and interview activities.



5.3 BACKEND IMPLEMENTATION

The backend of the system is created on Node.js and Express. Its functions include the processing of the main logic of the application, which includes such features as authentication, user management, resume processing, ATS analysis, job matching, interview processing, communication evaluation, coding, aptitude testing, skill recommendation, learning path recommendation, and performance management. The backend of the system has a modular structure with routes, controllers, services, middleware, and database models.



id	email	password_hash	name	college
00000000-0000-0000-0000-000000	demo@interviewcoach.ai	\$2a\$10\$demohashdemohashdemohas	Demo Student	EMPTY
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5.4 DATABASE IMPLEMENTATION

Supabase PostgreSQL is the database used to manage and store the application data. It uses a relational structure that arranges the data concerning users, resumes, interviews, aptitude test attempts, coding test attempts, communication tests, learning paths, and skill assessments.

Information about the user and the authentication process is managed and stored, while resume data stores the uploaded resume data and analysis. Information about interview questions and answers, their scores, and feedback is stored in the interview data. Aptitude test attempts and coding test attempts store assessment information and the performance data. Communication tests store communication evaluation data. Learning path and skill assessment data store the personalized learning data, identified skill gaps, recommendations, and progress.

5.5 AI-BASED RESUME AND INTERVIEW ANALYSIS

The AI-powered analysis tool will help in resume evaluation and interview feedback in the system. The AI will use internal analysis algorithms and, optionally, use external Large Language Models (LLM) to provide more comprehensive analysis. In resume analysis, the system finds important data, skills, and keywords related to the uploaded resume. In interviews, the system analyses user answers depending on certain characteristics and provides the required feedback. It is necessary to note that AI in the system will help students prepare for placements but will not replace human recruiters and make decisions regarding recruiting people. As AI-based evaluation can have drawbacks like providing incorrect feedback or unfair evaluation, it can only be considered as a guide for improvement.

5.6 ATS AND JOB MATCHING IMPLEMENTATION

The ATS system provides a structured score based on multiple factors.

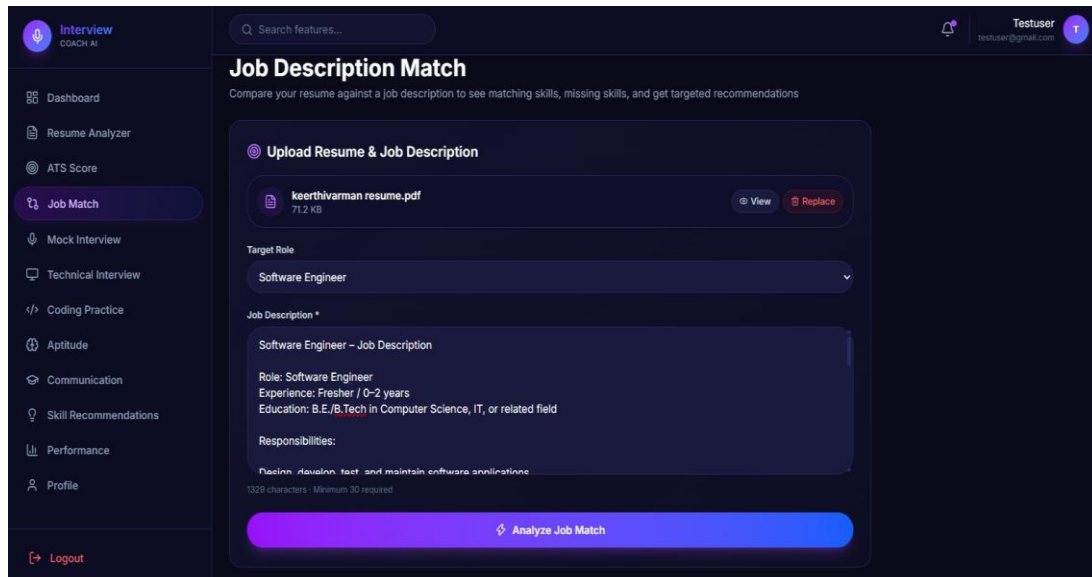
The scoring components are:

COMPONENT	MAXIMUM SCORE
Keyword Match	35
Experience / Projects	20
Section Completeness	15
Structure	15
Terminology	10
Contact Information	5
TOTAL	100

The job matching system then compares the resume with a selected job description.

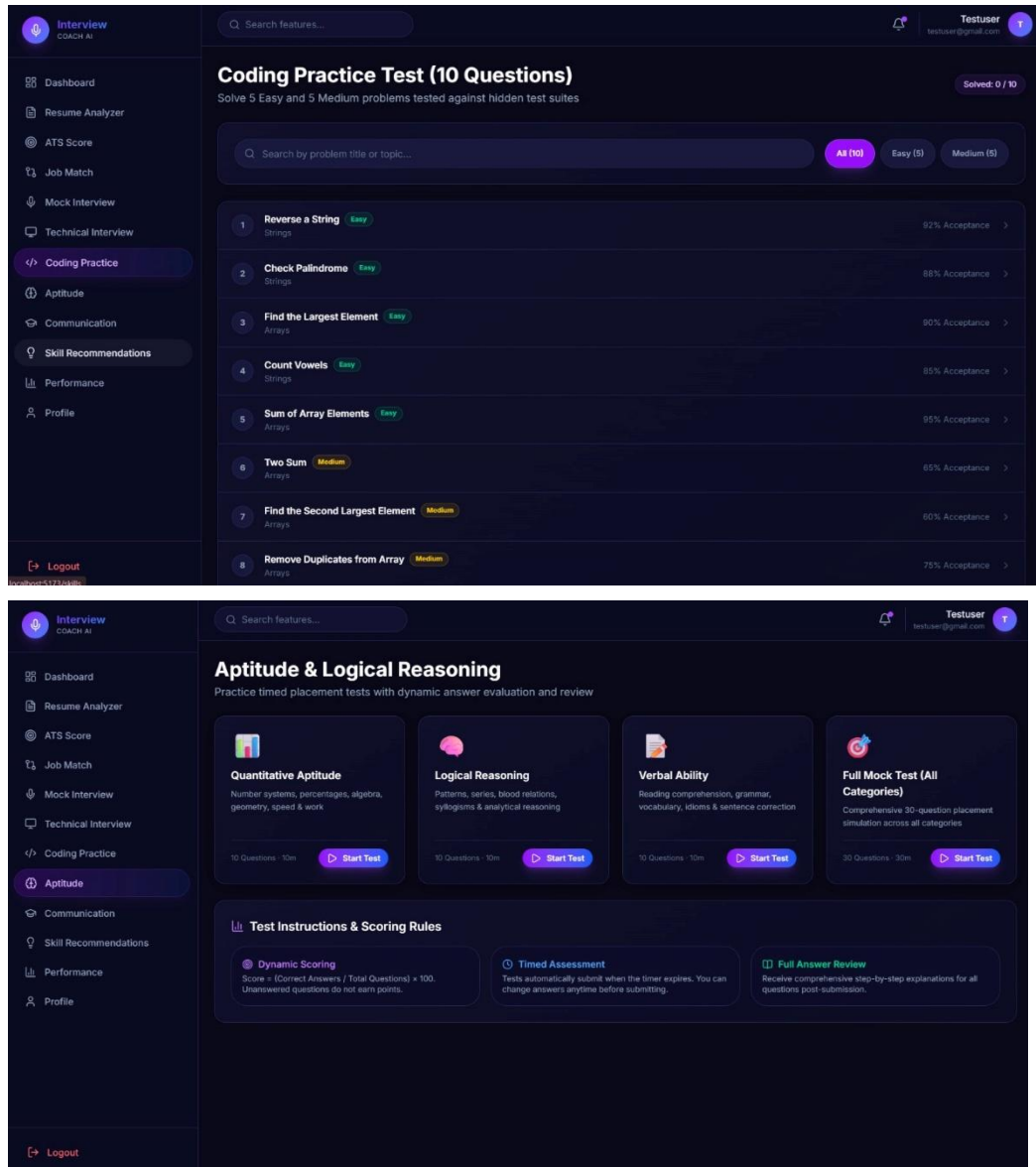
The result provides an understandable representation of how closely the candidate's current resume matches the selected role.

The screenshot displays the 'Interview COACH AI' dashboard. On the left is a sidebar menu with options: Dashboard, Resume Analyzer, ATS Score (highlighted), Job Match, Mock Interview, Technical Interview, Coding Practice, Aptitude, Communication, Skill Recommendations, Performance, and Profile. At the bottom of the sidebar is a 'Logout' button. The main content area is titled 'Role-Specific ATS Score' with the subtitle 'Evaluate how well your resume matches Applicant Tracking Systems for your target technical role'. Below this is a section 'Upload Resume & Select Target Role'. It features a large dashed box for uploading a resume file, with instructions 'Drag & drop your resume file' and 'or click to select file from your computer'. A note specifies 'Supported Formats: PDF, DOC, DOCX (Max 10MB)'. Below the upload area is a dropdown menu for 'Selected Technical Role *', which currently shows 'Software Engineer'. The top right of the dashboard shows a search bar, a notification bell, and a user profile for 'Testuser' with the email 'testuser@gmail.com'.



5.7 CODING AND APTITUDE ASSESSMENT

The Coding section enables students to solve programming problems and run their solution as part of their placements preparation. Information on the language used, problem chosen, code, submission status, score, number of passed test cases, total test cases, and execution time is recorded in the system. The Aptitude section tracks the performance in the aptitude test, which includes information on score, accuracy, time spent, and weaknesses identified. Information gathered in the coding and aptitude tests will enable the students to analyze their performance and identify areas needing improvement.

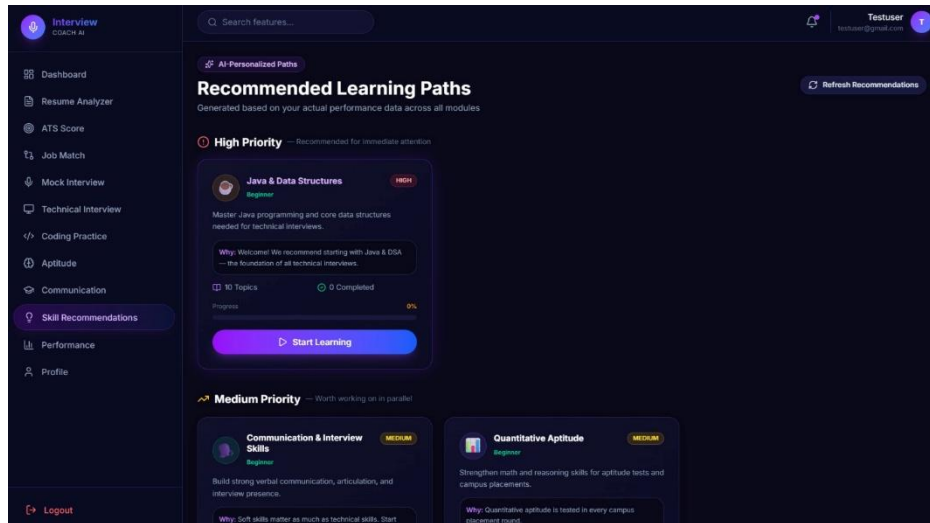


5.8 PERSONALIZED RECOMMENDATION AND LEARNING PATH

The recommendation system relies on the performance information stored for that user to determine the areas that need improvement.

Rather than offering preparation tips to everyone, the system tries to link

the recommendation with the performance of that particular individual.



5.9 AUTHENTICATION AND SECURITY

Security is one of the key elements of the application as the system will be processing user profiles, resumes, evaluation results, and authentication data. In order to secure user accounts in the system, password hashing and JWT authentication techniques are employed, while protected API routes and authentication middleware are used in order to limit access only to authenticated users. Rate limiting and security headers are implemented in order to increase the level of security for the application. File validation and size limits allow handling uploaded resumes in a safe manner. Configuration based on environment helps to protect sensitive parameters, while user ownership verification helps to avoid any unauthorized access to the other user's data.

5.10 ADMIN/STAFF PORTAL IMPLEMENTATION

The Admin/Staff Portal is created in a separate portal which is used by authorized staff members only. There is a portal selection interface provided in the application where a user is asked whether he/she wants to access the Student Portal or Staff Portal. Staff members should be logged in using their authorized logins before gaining access to the staff features. The Staff Portal offers facilities to monitor the performance of students, review interviews conducted and get the analytics. The separate staff authentication ensures that administrative portals are not accessible by unauthorized users.

5.11 RESULTS DISCUSSION

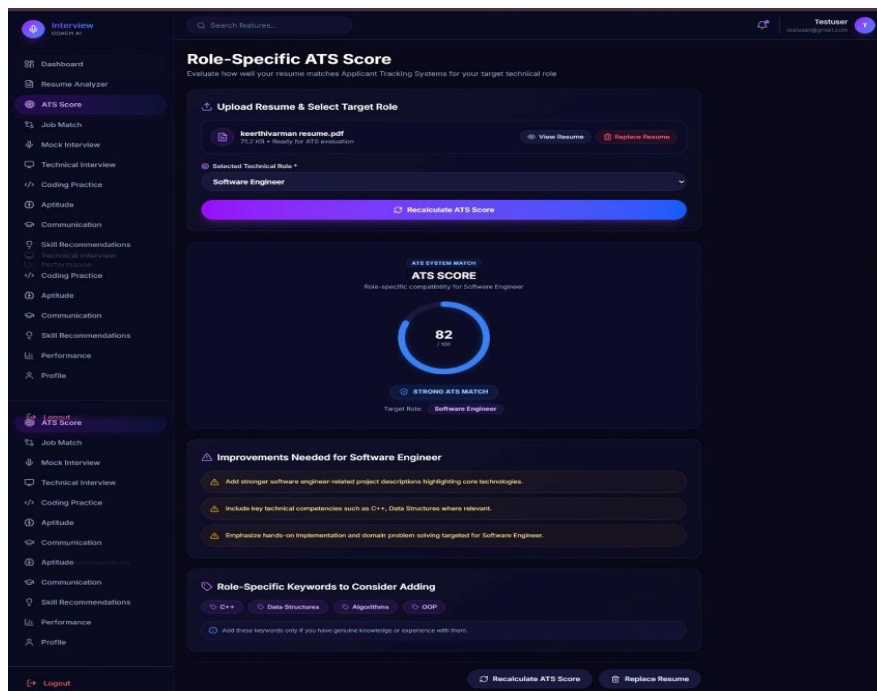
RESUME ANALYSIS RESULT

The Resume Analyzer conducts a comprehensive analysis of the uploaded resume in order to help the user understand how good it is as a whole, what should be added, which sections are missing, and how to improve the resume. The tool also highlights keywords that are missing from the resume and points to the problems of resume quality.



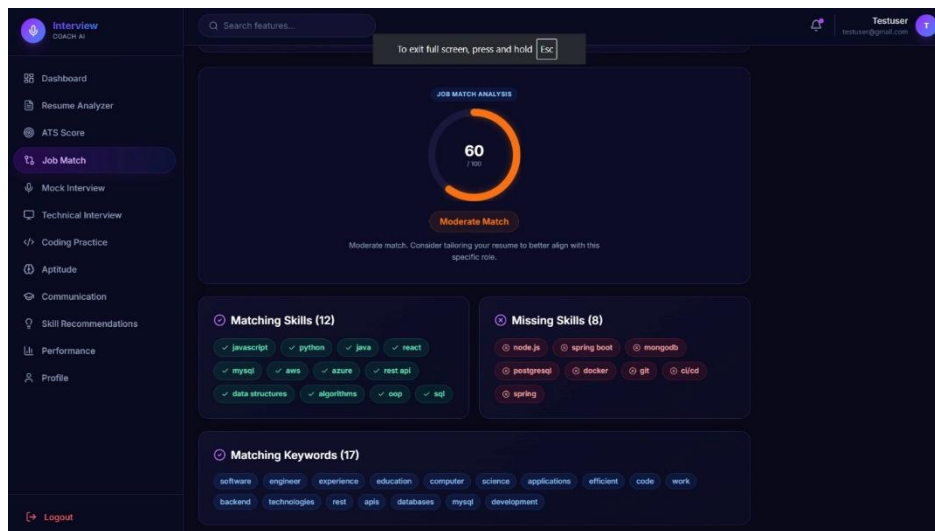
ATS SCORE RESULT

The result obtained by the ATS module is a score out of 100. This score shows users the elements of their resume that are responsible for the general ATS score.



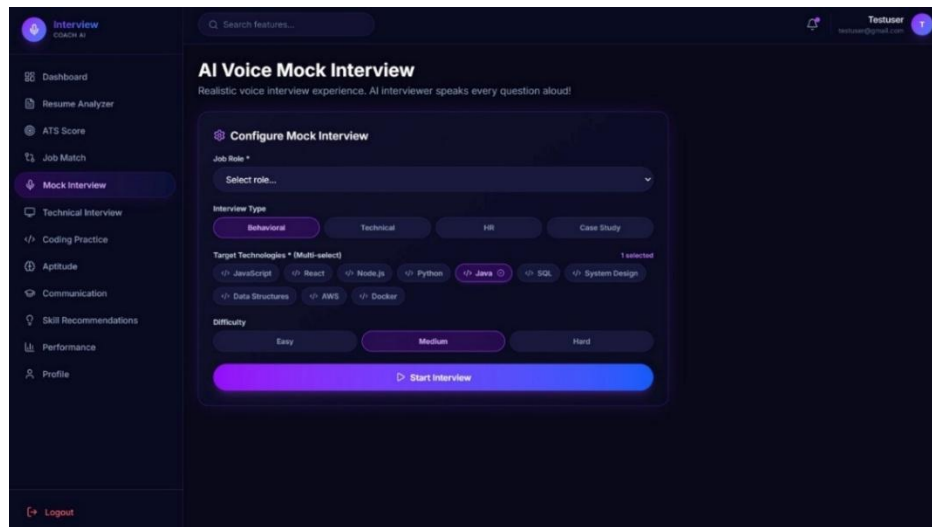
JOB MATCHING RESULT

The Job Matching module is aimed at identifying the connection between the resume of the candidate and the job description selected. It highlights matching skills and requirements that are still not included in the resume



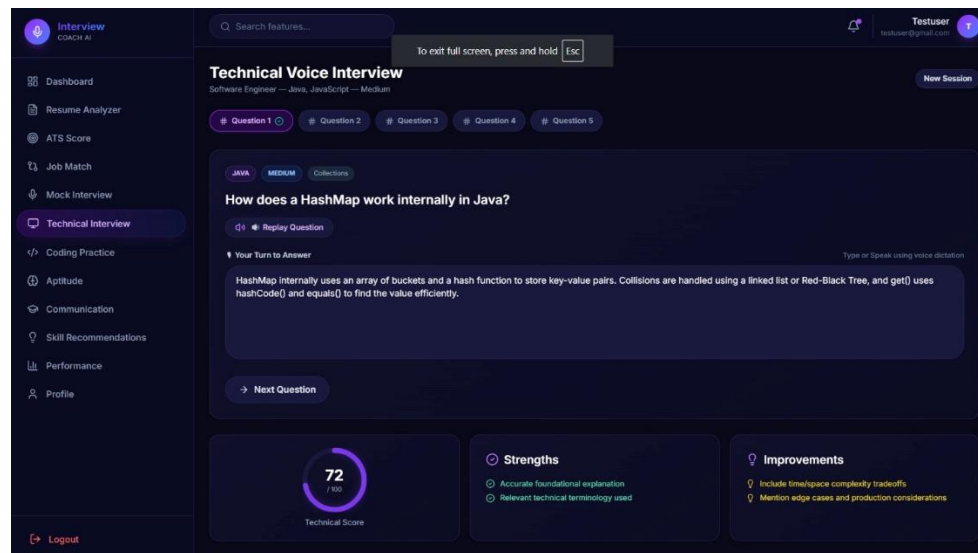
AI MOCK INTERVIEW RESULT

The mock interview module generates role-specific interview questions and evaluates user's answers to those. The user can analyze this feedback.



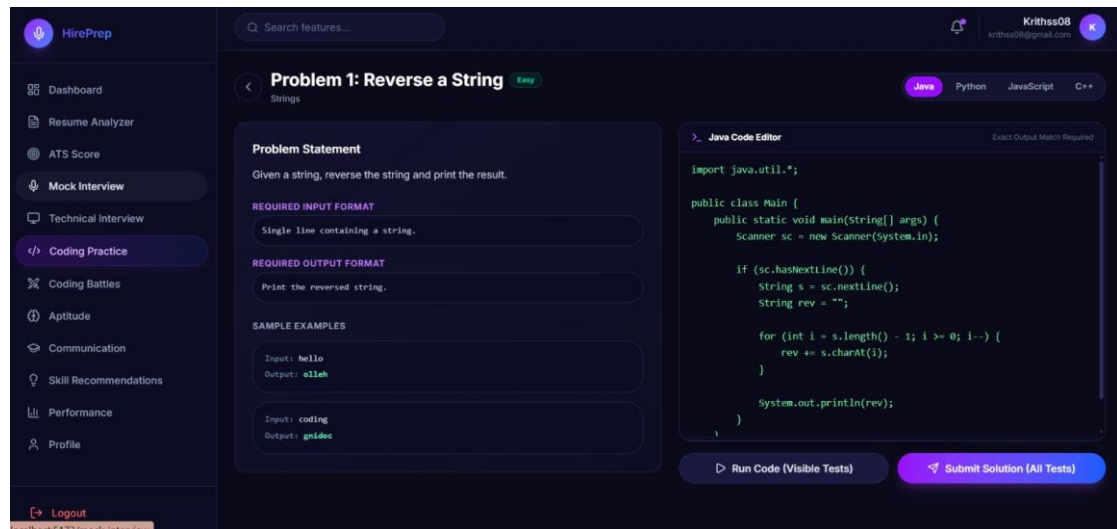
TECHNICAL INTERVIEW RESULT

The technical interview component evaluates the user's technical responses. It provides an opportunity to practice technical interview questions before attending an actual interview.



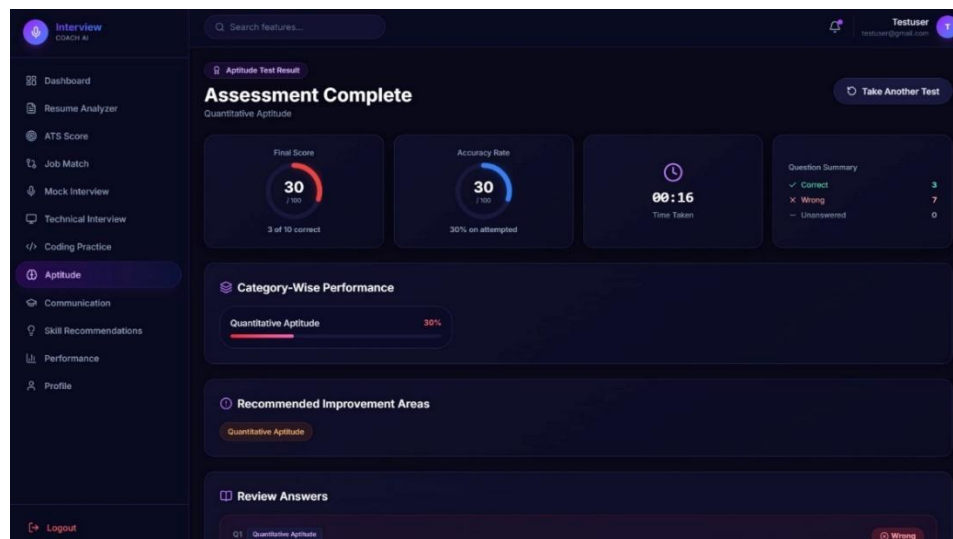
CODING PRACTICE RESULT

The coding module allows users to practice programming questions and test their solutions. The results help users understand their coding performance and identify areas that need additional practice.



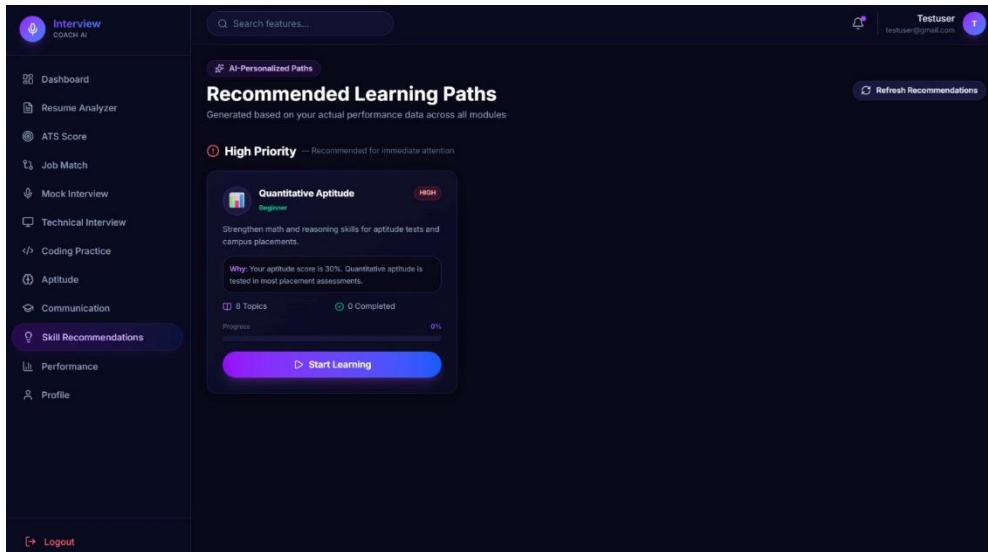
APTITUDE ASSESSMENT RESULT

The aptitude module provides performance information based on the completed assessment. The result can be used to identify weak areas and improve preparation.



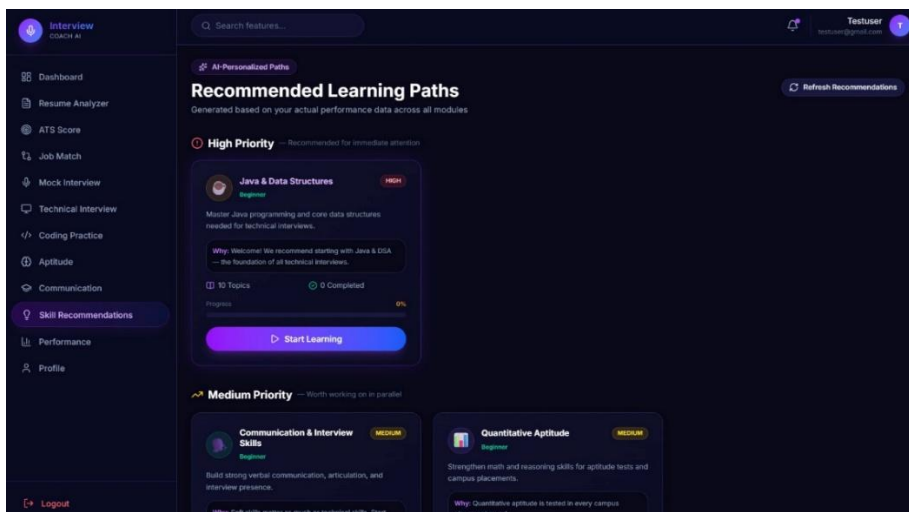
SKILL RECOMMENDATION RESULT

The skill recommendation module converts performance results into specific improvement suggestions. This helps users understand what they should focus on next.



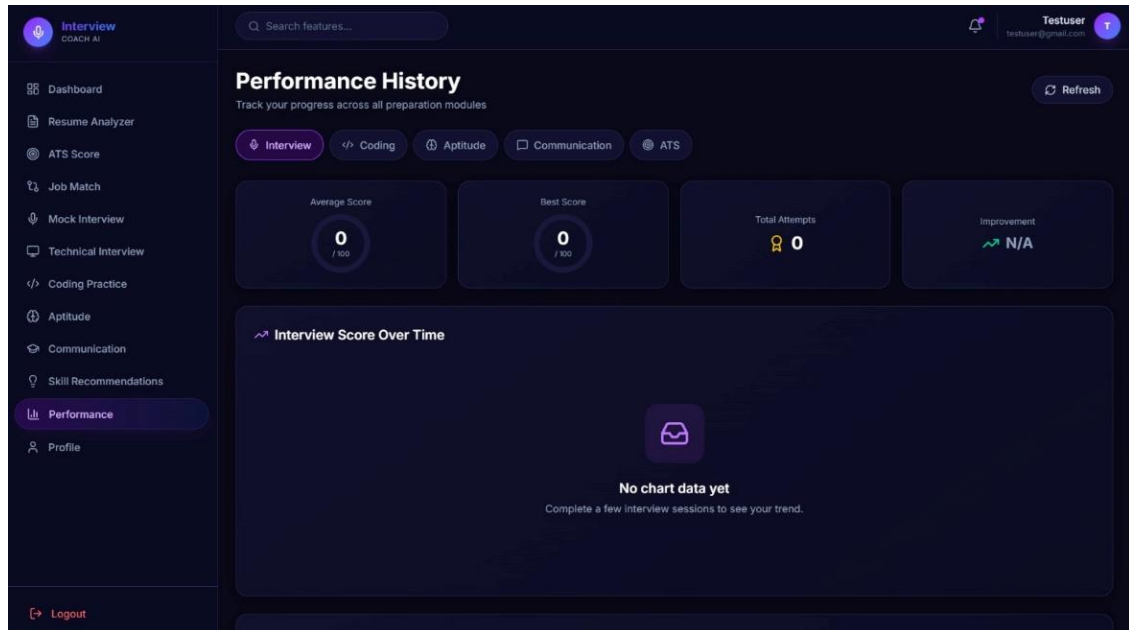
LEARNING PATH RESULT

The learning path provides a structured set of activities based on identified skill gaps. It allows users to work systematically on their weaker areas.



PERFORMANCE DASHBOARD RESULT

The performance dashboard combines information from multiple modules into one view. It provides an overall understanding of the student's preparation and progress.



CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

AI-Powered Interview Preparation & Resume Coach is a platform that will enable students to engage in all activities required during placements. The platform will facilitate the integration of important placement activities such as resume analysis, resume assessment, job matching, AI interview, technical preparation, communication evaluation, coding, aptitude evaluation, skill recommendations, learning path, and performance assessment in one platform.

Through the integration, the platform will help students to analyze their resumes, discover missing job requirements, practice interview questions, enhance their communication skills, improve their programming capabilities, evaluate their aptitude performance, and discover areas that need additional preparation. Through the combination of results generated from various activities, the platform will be able to make more relevant recommendations instead of evaluating each assessment activity separately.

In essence, the project creates an opportunity for students to continue preparing, assessing their performance, discovering their weaknesses, and working on becoming better and job-ready. The platform will solve the problem of scattered placement preparation and personalized recommendations through the consolidation of these activities in one platform.

6.2 FUTURE WORK

Improvement of the current platform can include incorporation of advanced functionalities to enable the platform become more intelligent and helpful for placement preparation. Further development versions of the project can include incorporation of more advanced LLMs to enable better understanding of the resumes and answers during interviews. Advanced voice analysis can be included to analyze speaking speed, pauses, pronunciation, usage of filler words, voice modulation, etc. Computer vision technologies can be used to analyze eye contact, facial expressions, posture, etc., during video interviews.

The job matching engine can be developed to incorporate semantic-based matching using embeddings for better understanding of skills and job requirements not only by keywords but also by their meanings. The coding engine can be further improved to support more programming languages and diverse problems. Finally, the platform can give users advanced job recommendations taking into account skills, resumes data, preferred job types, performance results, etc.

The interview component can also be improved using adaptive questioning which entails variation of questions' types and level of difficulty depending on the previous answers by the user. An app can further be designed for Android and iOS for mobile access to the system. Lastly, the feature of performance prediction can be added to analyze the overall performance of the student for placement purposes.